

Orion68–CPU_cmos

File: orion68cpu2.kicad_sch

Orion68–MEM

File: orion68mem.kicad_sch

Orion68–IO

File: orion68io.kicad_sch

Orion68–VID

File: orion68vid.kicad_sch

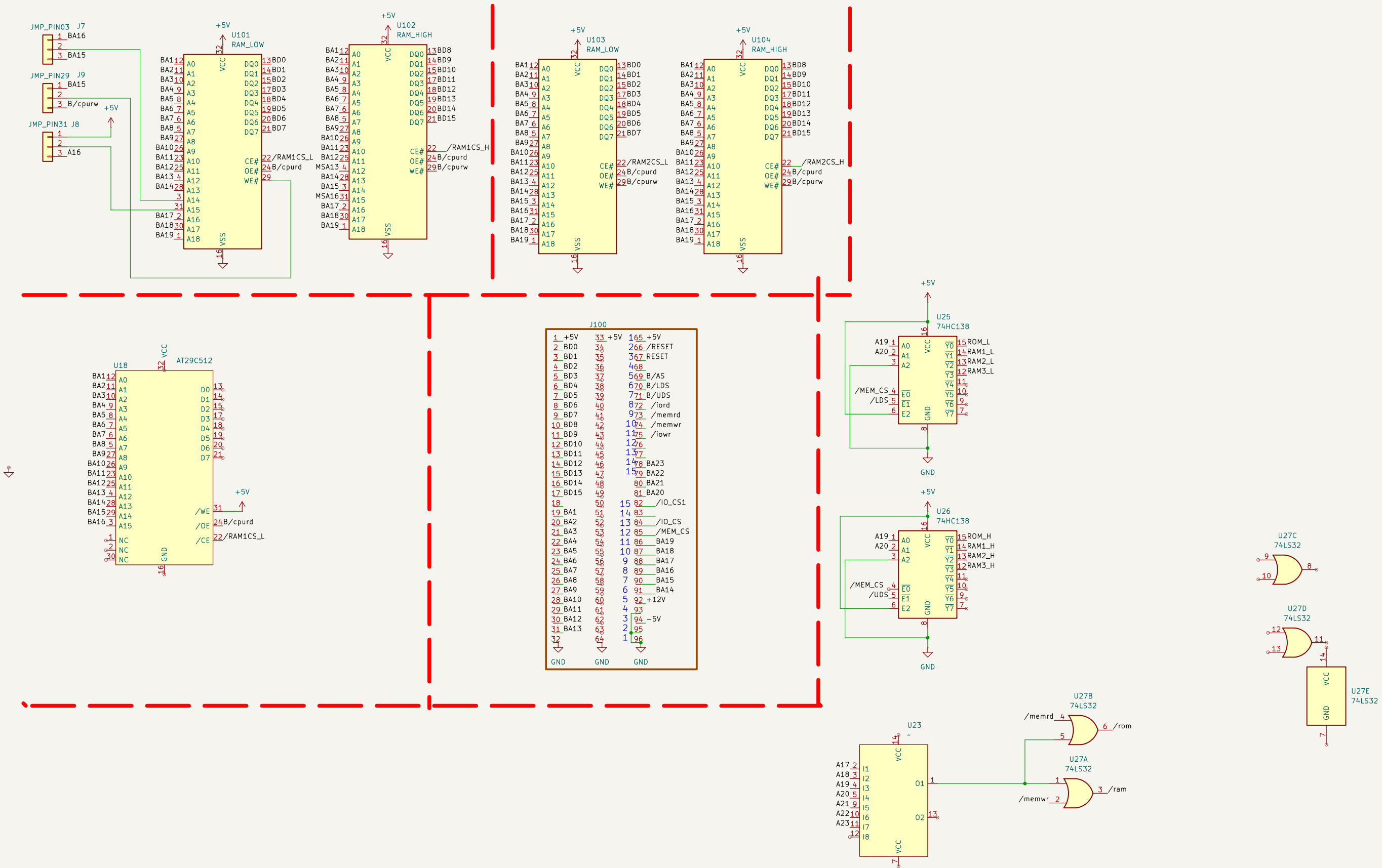
Orion68–CPU_nmos

File: cpu2.kicad_sch

Orion68–serial

File: orion68_serial.kicad_sch

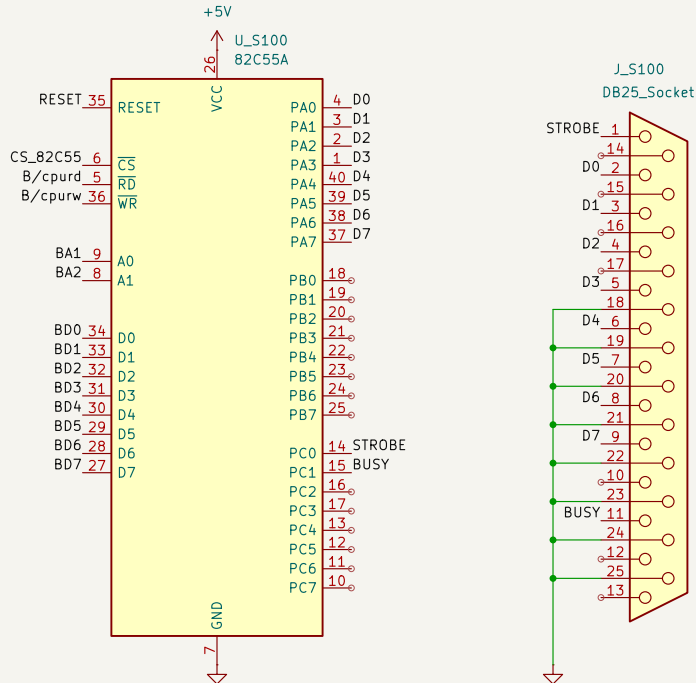
J4		
1 +5V	33 +5V	65 +5V
2 BD0	34 E	66 /RESET
3 BD1	35 CRUCLK	67 RESET
4 BD2	36 CRUIN	68 /BERR
5 BD3	37 CRUOUT	69 Res9
6 BD4	38 DBIN	70 B/LDS
7 BD5	39 IAQ	71 B/UDS
8 BD6	40 /INTR	72 /INTA
9 BD7	41 NMI	73 ALE
10 BD8	42 /HOLD	74 /DEN
11 BD9	43 B/AS	75 DT/R
12 BD10	44 /iord	76 M/IO
13 BD11	45 /memrd	77 /SSO
14 BD12	46 /memwr	78 /WAIT
15 BD13	47 /iowr	79 /M1
16 BD14	48 IRQ2	80 /RFSH
17 BD15	49 IRQ3	81 CPU_CLK
18 BA0	50 IRQ4	82 /IO_CS1
19 BA1	51 /RST	83 /UART_CS
20 BA2	52 HLDA	84 /IO_CS
21 BA3	53 /HALT	85 /MEM_CS
22 BA4	54 BA23	86 TICK_IN
23 BA5	55 BA22	87 BOOTIN
24 BA6	56 BA21	88 PWR_GOOD
25 BA7	57 BA20	89 IRQ5
26 BA8	58 BA19	90 IRQ6
27 BA9	59 BA18	91 IRQ7
28 BA10	60 BA17	92 +12V
29 BA11	61 BA16	93
30 BA12	62 BA15	94 -5V
31 BA13	63 BA14	95
32	64	96
GND	GND	GND



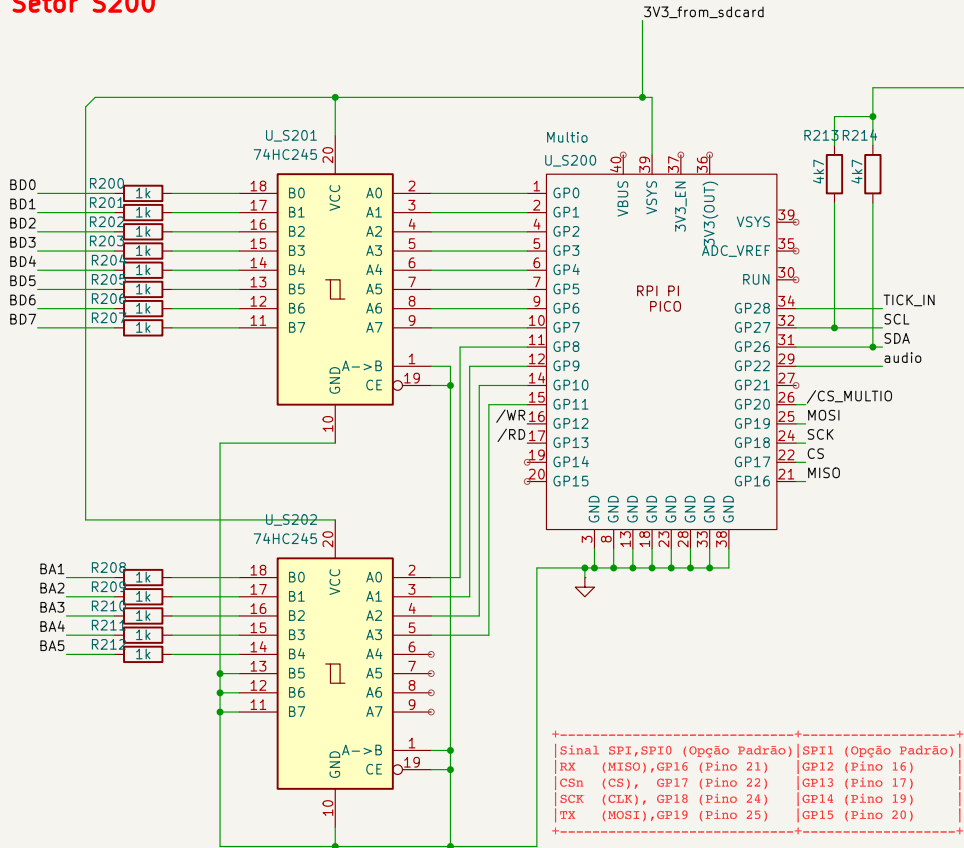
Memory board - 100

Setor S100

DB25 fêmea			
Pino	Nome	Direção (PC)	Função
1	,Strobe	,Saída	,Indica que os dados são válidos (Ativo em 0)
2	,D0	,E/S	,Bit de Dados 0 (LSB)
3	,D1	,E/S	,Bit de Dados 1
4	,D2	,E/S	,Bit de Dados 2
5	,D3	,E/S	,Bit de Dados 3
6	,D4	,E/S	,Bit de Dados 4
7	,D5	,E/S	,Bit de Dados 5
8	,D6	,E/S	,Bit de Dados 6
9	,D7	,E/S	,Bit de Dados 7 (MSB)
10	,Ack	,Entrada	,Reconhecimento de recebimento (Ativo em 0)
11	,Busy	,Entrada	,Impressora ocupada
12	,Paper Out	,Entrada	,Fim de papel
13	,Select	,Entrada	,Impressora online
14	,Auto Linefeed	,Saída	,Avanço de linha automático
15	,Error	,Entrada	,Erro na impressora
16	,Init	,Saída	,Inicializar impressora
17	,Select In	,Saída	,Selecionar impressora
18-25	,GND	,---	,Terra (Ground)



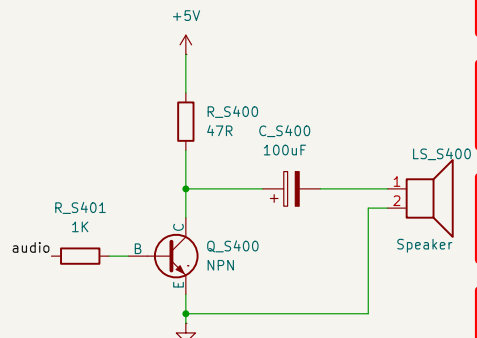
Setor S200



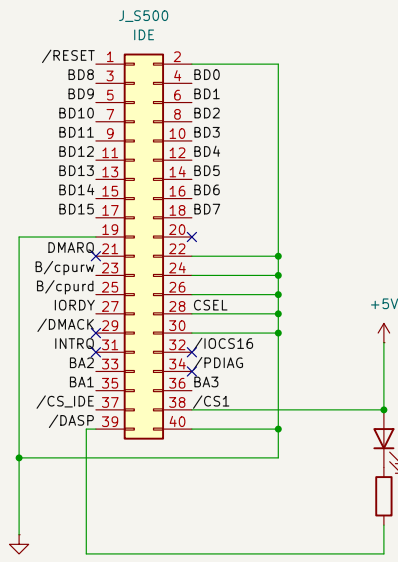
Setor S400



0xF00040-0xF0007F -> 82C55
0xF00080-0xF000BF -> IDE
0xF000C0-0xF000FF -> RTC
0xF00140-0xF0017F -> TIMER
0xF00180-0xF001BF -> SDCARD
0xF001C0-0xF001FF -> free
0xF00240-0xF0027F -> free
0xF00280-0xF002BF -> free
0xF002C0-0xF002FF -> free
0xF00340-0xF0037F -> free
0xF00380-0xF003BF -> free
0xF003C0-0xF003FF -> free
0xF00400-0xF0043F -> free
0xF00440-0xF0047F -> free
0xF00480-0xF004BF -> free
0xF004C0-0xF004FF -> free
0xF00500-0xF0053F -> free
0xF00540-0xF0057F -> free
0xF00580-0xF005BF -> free
0xF005C0-0xF005FF -> free
0xF00600-0xF0063F -> free
0xF00640-0xF0067F -> free
0xF00680-0xF006BF -> free
0xF006C0-0xF006FF -> free
0xF00700-0xF0073F -> free
0xF00740-0xF0077F -> free
0xF00780-0xF007BF -> free
0xF007C0-0xF007FF -> free
0xF00800-0xF0083F -> free
0xF00840-0xF0087F -> free
0xF00880-0xF008BF -> free
0xF008C0-0xF008FF -> free
0xF00900-0xF0093F -> free
0xF00940-0xF0097F -> free
0xF00980-0xF009BF -> free
0xF009C0-0xF009FF -> free
0xF00A00-0xF00A3F -> free
0xF00A40-0xF00A7F -> free
0xF00A80-0xF00ABF -> free
0xF00AC0-0xF00AFF -> free
0xF00B00-0xF00B3F -> free
0xF00B40-0xF00B7F -> free
0xF00B80-0xF00BBF -> free
0xF00BC0-0xF00BFF -> free
0xF00C00-0xF00C3F -> free
0xF00C40-0xF00C7F -> free
0xF00C80-0xF00CBF -> free
0xF00CC0-0xF00CFF -> free
0xF00D00-0xF00D3F -> free
0xF00D40-0xF00D7F -> free
0xF00D80-0xF00DBF -> free
0xF00DC0-0xF00DFF -> free
0xF00E00-0xF00E3F -> free
0xF00E40-0xF00E7F -> free
0xF00E80-0xF00EBF -> free
0xF00EC0-0xF00EFF -> free
0xF00F00-0xF00F3F -> free
0xF00F40-0xF00F7F -> free
0xF00F80-0xF00FBF -> free
0xF00FC0-0xF00FFF -> free



Setor S500



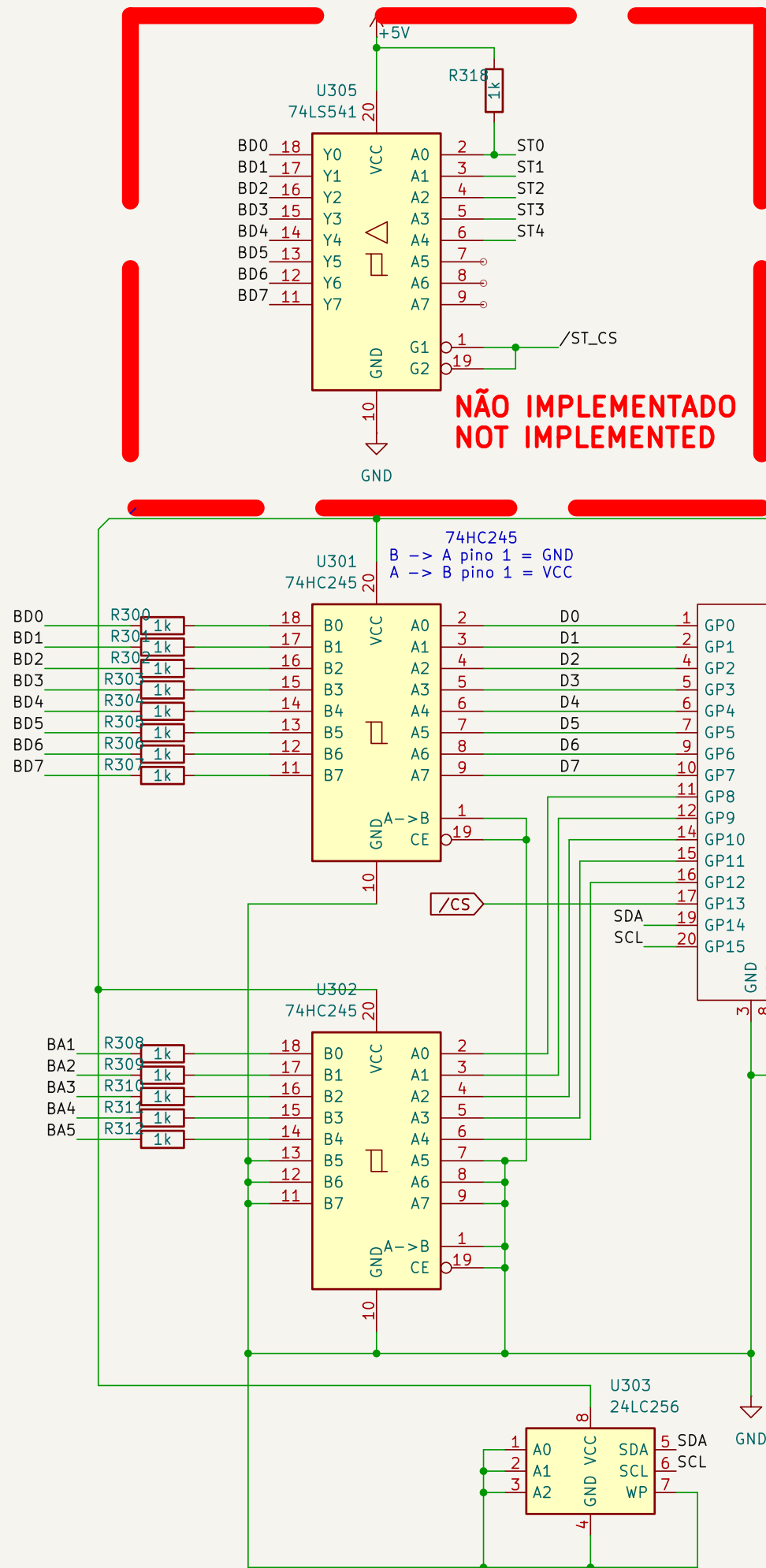
Binário	LDS	Ender.	D0-D7 (LDS)
A5...A1	A0	Decimal	D8-D15 (UDS)
00000	0	0 (Par)	LDS
00000	1	1 (Ímpar)	UDS
00001	0	2 (Par)	LDS
00001	1	3 (Ímpar)	UDS
00010	0	4 (Par)	LDS
00010	1	5 (Ímpar)	UDS
00011	0	6 (Par)	LDS
00011	1	7 (Ímpar)	UDS
00100	0	8 (Par)	LDS
00100	1	9 (Ímpar)	UDS
00101	0	10 (Par)	LDS
00101	1	11 (Ímpar)	UDS
00110	0	12 (Par)	LDS
00110	1	13 (Ímpar)	UDS
00111	0	14 (Par)	LDS
00111	1	15 (Ímpar)	UDS
01000	0	16 (Par)	LDS
01000	1	17 (Ímpar)	UDS
01001	0	18 (Par)	LDS
01001	1	19 (Ímpar)	UDS
01010	0	20 (Par)	LDS
01010	1	21 (Ímpar)	UDS
01011	0	22 (Par)	LDS
01011	1	23 (Ímpar)	UDS
01100	0	24 (Par)	LDS
01100	1	25 (Ímpar)	UDS
01101	0	26 (Par)	LDS
01101	1	27 (Ímpar)	UDS
01110	0	28 (Par)	LDS
01110	1	29 (Ímpar)	UDS
01111	0	30 (Par)	LDS
01111	1	31 (Ímpar)	UDS
10000	0	32 (Par)	LDS
10000	1	33 (Ímpar)	UDS
10001	0	34 (Par)	LDS
10001	1	35 (Ímpar)	UDS
10010	0	36 (Par)	LDS
10010	1	37 (Ímpar)	UDS
10011	0	38 (Par)	LDS
10011	1	39 (Ímpar)	UDS
10100	0	40 (Par)	LDS
10100	1	41 (Ímpar)	UDS
10101	0	42 (Par)	LDS
10101	1	43 (Ímpar)	UDS
10110	0	44 (Par)	LDS
10110	1	45 (Ímpar)	UDS
10111	0	46 (Par)	LDS
10111	1	47 (Ímpar)	UDS
11000	0	48 (Par)	LDS
11000	1	49 (Ímpar)	UDS
11001	0	50 (Par)	LDS
11001	1	51 (Ímpar)	UDS
11010	0	52 (Par)	LDS
11010	1	53 (Ímpar)	UDS
11011	0	54 (Par)	LDS
11011	1	55 (Ímpar)	UDS
11100	0	56 (Par)	LDS
11100	1	57 (Ímpar)	UDS
11101	0	58 (Par)	LDS
11101	1	59 (Ímpar)	UDS
11110	0	60 (Par)	LDS
11110	1	61 (Ímpar)	UDS
11111	0	62 (Par)	LDS
11111	1	63 (Ímpar)	UDS

J200

1 +5V	33 +5V	65 +5V
2 BD0	34 E	66 /RESET
3 BD1	35 CRUCL	67 RESET
4 BD2	36 CRUIN	68 /BERR
5 BD3	37 CRUOU	69 Res9
6 BD4	38 DBIN	70 B/LDS
7 BD5	39 IAQ	71 B/UDS
8 BD6	40 /INTR	72 /INTA
9 BD7	41 NMI	73 ALE
10 BD8	42 /HOLD	74 /DEN
11 BD9	43 B/AS	75 DT/R
12 BD10	44 RDY	76 M/IO
13 BD11	45 B/cpurw	77 /SSO
14 BD12	46 B/cpurw	78 /WAIT
15 BD13	47 IC3	79 /M1
16 BD14	48 IRQ2	80 /RFSH
17 BD15	49 IRQ3	81 CPU_CLK
18 BA0	50 IRQ4	82
19 BA1	51 /RST	83
20 BA2	52 HLDA	84
21 BA3	53 /HALT	85 /IO_CS
22 BA4	54 BA23	86 TICK_IN
23 BA5	55 BA22	87 BOOTIN
24 BA6	56 BA21	88 PWR_GOOD
25 BA7	57 BA20	89 IRQ5
26 BA8	58 BA19	90 IRQ6
27 BA9	59 BA18	91 IRQ7
28 BA10	60 BA17	92 +12V
29 BA11	61 BA16	93
30 BA12	62 BA15	94 -5V
31 BA13	63 BA14	95
32	64	96
GND	GND	GND

IO board - 200

Orion68-IO

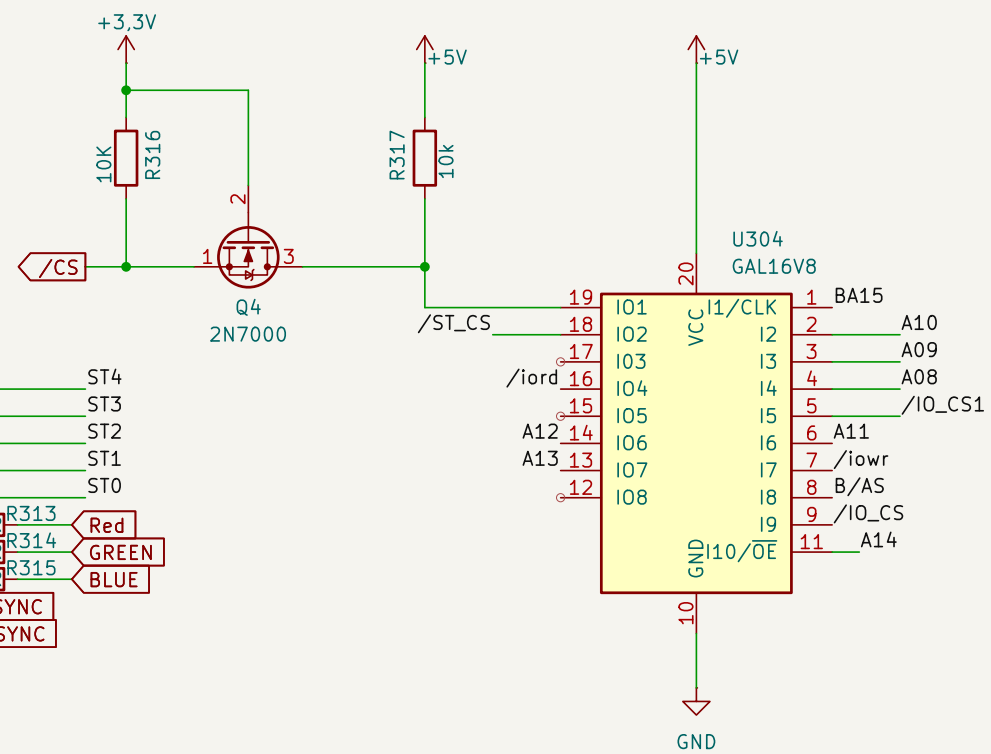


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19 18 17 16 | 15
1 0 0 0 | 0 = F80000
1 0 0 0 | 1 = F88000
1 0 0 1 | 0 = F90000
1 0 0 1 | 1 = F98000
1 0 1 0 | 0 = FA0000
1 0 1 0 | 1 = FA8000
1 0 1 1 | 0 = FB0000
1 0 1 1 | 1 = FB8000
pino 1 (A0) = A15 do 68000
pino 2 (A1) = A16 do 68000
pino 3 (A2) = A17 do 68000
pino 4 (/E0) = A18 do 68000
pino 5 (/E1) = /LDS R/W /AS do 68000
pino 6 (E2) = A19 do 68000

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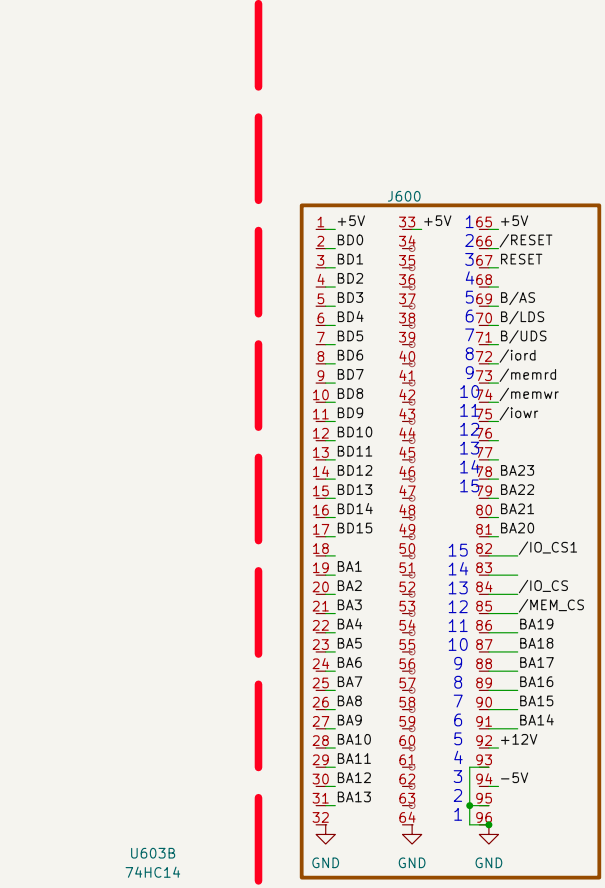
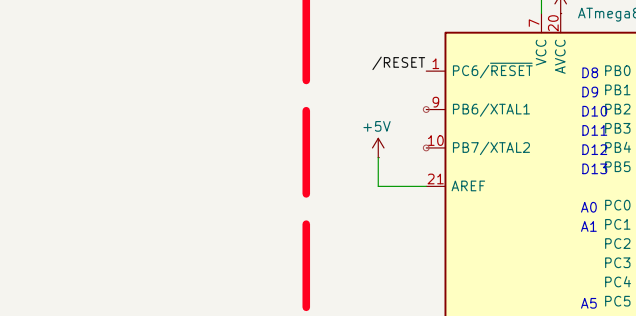
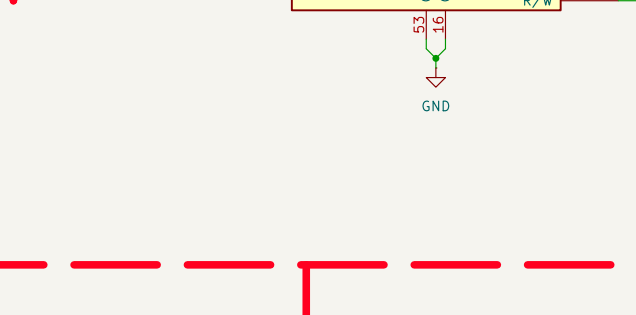
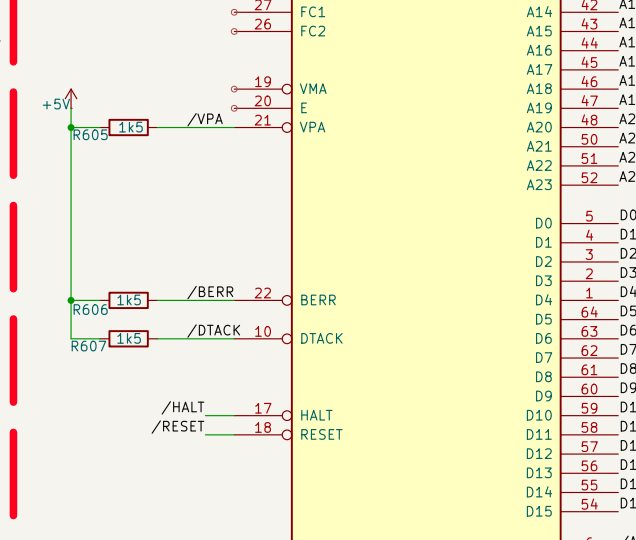
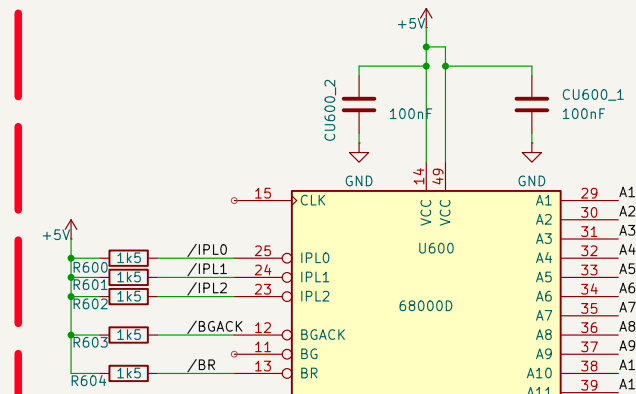
com isso a saída pino 7 /Y7 só deve ser acessada com 0xB8000



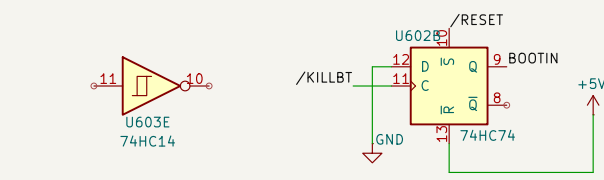
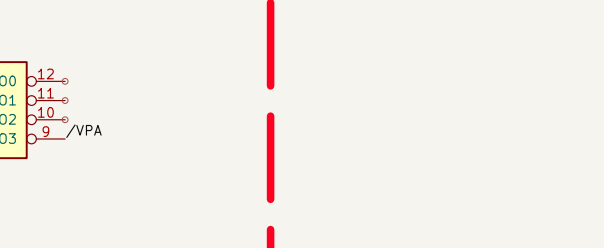
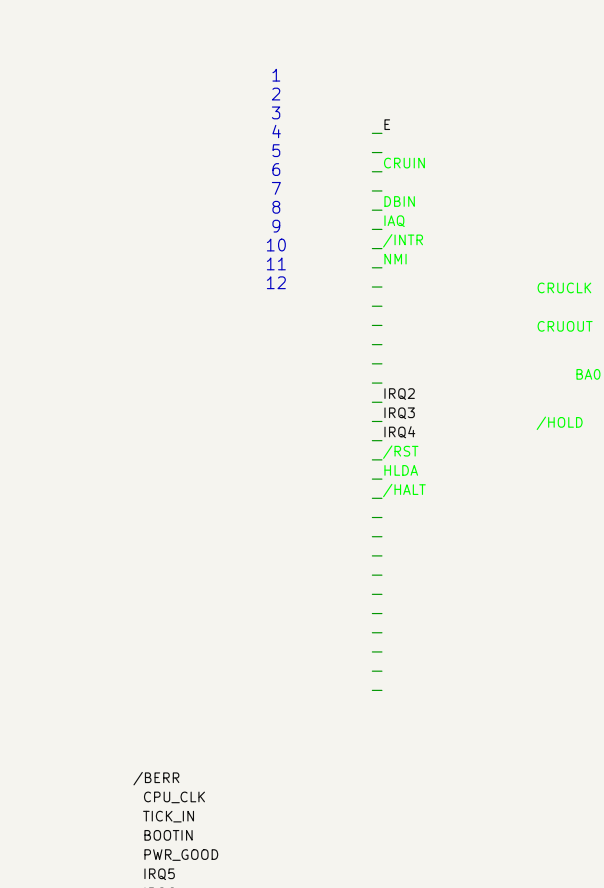
J300		
1 +5V	33 +5V	165 +5V
2 BD0	34	266 /RESET
3 BD1	35	367 RESET
4 BD2	36	468
5 BD3	37	569 B/AS
6 BD4	38	670 B/LDS
7 BD5	39	771 B/UDS
8 BD6	40	872 /iord
9 BD7	41	973 /memrd
10 BD8	42	1074 /memwr
11 BD9	43	1175 /iowr
12 BD10	44	1276
13 BD11	45	1377
14 BD12	46	1478 BA23
15 BD13	47	1579 BA22
16 BD14	48	80 BA21
17 BD15	49	81 BA20
18	50	15 82 /IO_CS1
19 BA1	51	14 83
20 BA2	52	13 84 /IO_CS
21 BA3	53	12 85 /MEM_CS
22 BA4	54	11 86 BA19
23 BA5	55	10 87 BA18
24 BA6	56	9 88 BA17
25 BA7	57	8 89 BA16
26 BA8	58	7 90 BA15
27 BA9	59	6 91 BA14
28 BA10	60	5 92 +12V
29 BA11	61	4 93
30 BA12	62	3 94 -5V
31 BA13	63	2 95
32	64	1 96
GND	GND	GND

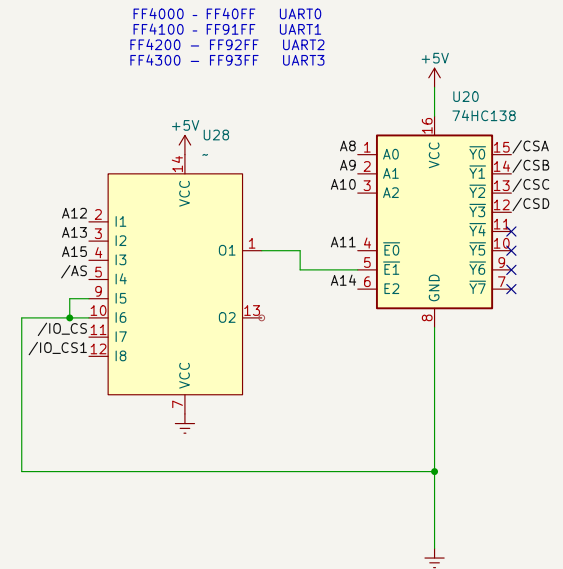
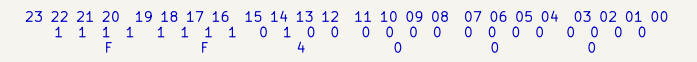
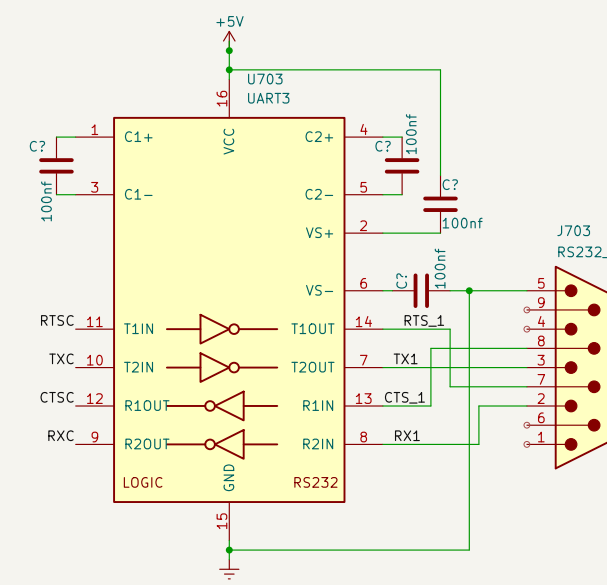
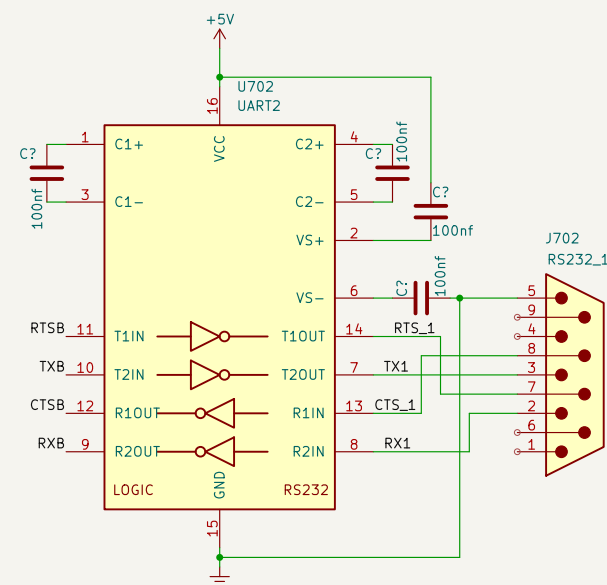
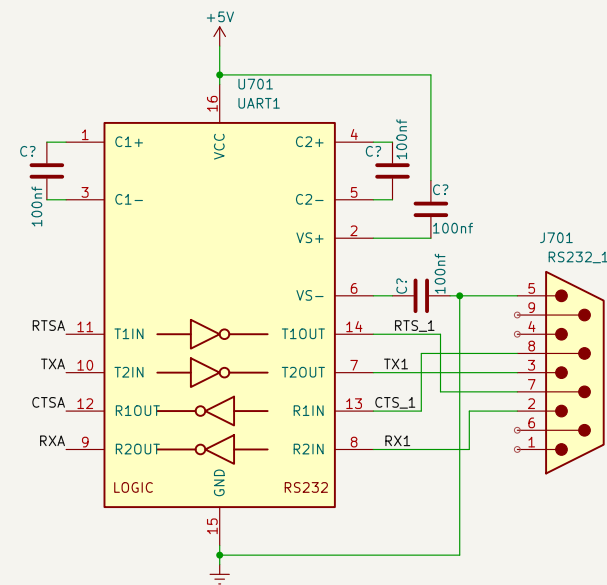
Orion68-VID

2026-03-14



J600		
<u>1</u> +5V	<u>33</u> +5V	<u>165</u> +5V
<u>2</u> BD0	<u>34</u>	<u>266</u> /RESET
<u>3</u> BD1	<u>35</u>	<u>367</u> /RESET
<u>4</u> BD2	<u>36</u>	<u>468</u>
<u>5</u> BD3	<u>37</u>	<u>569</u> B/AS
<u>6</u> BD4	<u>38</u>	<u>670</u> B/LDS
<u>7</u> BD5	<u>39</u>	<u>771</u> B/UDS
<u>8</u> BD6	<u>40</u>	<u>872</u> /iord
<u>9</u> BD7	<u>41</u>	<u>973</u> /memrd
<u>10</u> BD8	<u>42</u>	<u>1074</u> /memwr
<u>11</u> BD9	<u>43</u>	<u>1175</u> /lowr
<u>12</u> BD10	<u>44</u>	<u>1276</u>
<u>13</u> BD11	<u>45</u>	<u>1377</u>
<u>14</u> BD12	<u>46</u>	<u>1478</u> BA23
<u>15</u> BD13	<u>47</u>	<u>1579</u> BA22
<u>16</u> BD14	<u>48</u>	<u>80</u> BA21
<u>17</u> BD15	<u>49</u>	<u>61</u> BA20
<u>18</u>	<u>50</u>	<u>15</u> <u>82</u> /IO_CS1
<u>19</u> BA1	<u>51</u>	<u>14</u> <u>83</u>
<u>20</u> BA2	<u>52</u>	<u>13</u> <u>84</u> /IO_CS
<u>21</u> BA3	<u>53</u>	<u>12</u> <u>85</u> /MEM_CS
<u>22</u> BA4	<u>54</u>	<u>11</u> <u>86</u> BA19
<u>23</u> BA5	<u>55</u>	<u>10</u> <u>87</u> BA18
<u>24</u> BA6	<u>56</u>	<u>9</u> <u>88</u> BA17
<u>25</u> BA7	<u>57</u>	<u>8</u> <u>89</u> BA16
<u>26</u> BA8	<u>58</u>	<u>7</u> <u>90</u> BA15
<u>27</u> BA9	<u>59</u>	<u>6</u> <u>91</u> BA14
<u>28</u> BA10	<u>60</u>	<u>5</u> <u>92</u> +12V
<u>29</u> BA11	<u>61</u>	<u>4</u> <u>93</u>
<u>30</u> BA12	<u>62</u>	<u>3</u> <u>94</u> -5V
<u>31</u> BA13	<u>63</u>	<u>2</u> <u>95</u>
<u>32</u>	<u>64</u>	<u>1</u> <u>96</u>
		
GND	GND	GND





J700					
<u>1</u>	+5V	<u>33</u>	+5V	<u>165</u>	+5V
<u>2</u>	BD0	<u>34</u>		<u>266</u>	/RESET
<u>3</u>	BD1	<u>35</u>		<u>367</u>	RESET
<u>4</u>	BD2	<u>36</u>		<u>468</u>	
<u>5</u>	BD3	<u>37</u>		<u>569</u>	B/AS
<u>6</u>	BD4	<u>38</u>		<u>670</u>	B/LDS
<u>7</u>	BD5	<u>39</u>		<u>771</u>	B/UDS
<u>8</u>	BD6	<u>40</u>		<u>872</u>	/iord
<u>9</u>	BD7	<u>41</u>		<u>973</u>	/memrd
<u>10</u>	BD8	<u>42</u>		<u>1074</u>	/memwr
<u>11</u>	BD9	<u>43</u>		<u>1175</u>	/iowr
<u>12</u>	BD10	<u>44</u>		<u>1276</u>	
<u>13</u>	BD11	<u>45</u>		<u>1377</u>	
<u>14</u>	BD12	<u>46</u>		<u>1478</u>	BA23
<u>15</u>	BD13	<u>47</u>		<u>1579</u>	BA22
<u>16</u>	BD14	<u>48</u>		<u>80</u>	BA21
<u>17</u>	BD15	<u>49</u>		<u>81</u>	BA20
<u>18</u>		<u>50</u>		<u>1582</u>	/IO_CS1
<u>19</u>	BA1	<u>51</u>		<u>1483</u>	
<u>20</u>	BA2	<u>52</u>		<u>1384</u>	/IO_CS
<u>21</u>	BA3	<u>53</u>		<u>1285</u>	/MEM_CS
<u>22</u>	BA4	<u>54</u>		<u>1186</u>	BA19
<u>23</u>	BA5	<u>55</u>		<u>1087</u>	BA18
<u>24</u>	BA6	<u>56</u>		<u>988</u>	BA17
<u>25</u>	BA7	<u>57</u>		<u>889</u>	BA16
<u>26</u>	BA8	<u>58</u>		<u>790</u>	BA15
<u>27</u>	BA9	<u>59</u>		<u>691</u>	BA14
<u>28</u>	BA10	<u>60</u>		<u>592</u>	+12V
<u>29</u>	BA11	<u>61</u>		<u>493</u>	
<u>30</u>	BA12	<u>62</u>		<u>394</u>	-5V
<u>31</u>	BA13	<u>63</u>		<u>295</u>	
<u>32</u>		<u>64</u>		<u>196</u>	
		↓			
GND		GND		GND	

